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Ammonia-fueled solar combined cycle power system: Thermal and combustion efficiency enhancement via solar-driven ammonia decomposition

Dmitry Pashchenko *

Department of Mechanical Engineering (Robotics) and MATEC, Guangdong Technion–Israel Institute of Technology, 241 Daxue Road, Shantou, Guangdong, 515063, China

Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel

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ABSTRACT

Ammonia, derived from renewable energy sources, has garnered significant international attention as a carbon-free fuel for power generation. However, direct combustion of ammonia in gas turbines results in low combustion efficiency. Conversely, ammonia can be readily converted into a hydrogen-rich fuel that has higher combustion efficiency. In this paper, we studied the synergistic integration of ammonia-fed gas turbines and concentrated solar power plants, where solar energy is harnessed for the thermochemical decomposition of ammonia into hydrogen-rich gas. The hydrogen-rich gas produced on-site replaces ammonia as the fuel for the gas turbine, enhancing thermal and combustion efficiency. The study investigates the impact of operational parameters on thermal and combustion efficiency. It is established that using the proposed concept the thermal efficiency can be increased up to 3.5%. Furthermore, a thermodynamic analysis was conducted to assess the thermal efficiency and heat balance. The effect of operational parameters such as the composition of the hydrogen-rich gas, pressure, initial temperature, and fuel-air equivalence ratio, on flame speed is analyzed. The results reveals that under gas turbine operating conditions, the flame speed ranges from 20 to 30 cm/s for 1:3 mole-to-mole ammonia-hydrogen blend.

1. Introduction

In the quest for sustainable energy solutions, the focus on CO_2-free power generation has become extremely important [1–3]. The urgency to mitigate the impacts of climate change has driven the need to transition towards cleaner and more efficient power generation technologies [4,5]. Ammonia has emerged as a promising alternative fuel for gas turbines, offering a pathway towards decarbonization and reduced emissions [6,7]. As a hydrogen carrier, it can be produced from renewable sources such as wind, solar, or hydroelectric power through the Haber-Bosch process, offering a pathway for sustainable energy production [8,9]. When combusted, it mainly releases nitrogen and water vapor, minimizing environmental impact [10]. Furthermore, its existing infrastructure for storage and transportation, along with its compatibility with existing combustion technologies, makes it a viable alternative to traditional fossil fuels. However, the adoption of ammonia as a fuel source presents both opportunities and challenges that need to be addressed [11]. For example, if ammonia is used directly as gas turbine fuel, the gas turbine suffers from low combustion efficiency due to the low flame speed and long quenching distance of the fuel [12,13].

Another promising direction to carbon-free energy is the use of solar energy [14–16]. Concurrently, the utilization of solar energy in power plants has gained significant traction due to its abundant and renewable nature [17,18]. Concentrated solar power (CSP) plants have demonstrated (e.g. Ivanpah Solar Power Facility (USA); Noor Ouarzazate Complex (Morocco); Gemasolar Thermosolar Plant (Spain)) the potential to harness solar energy efficiently, providing a sustainable source of electricity [19–23]. Despite the advantages of solar energy, there are inherent challenges, such as intermittency and variability, that need to be overcome for widespread integration into the energy grid [24,25]. The synergy between ammonia-fired gas turbines and concentrated solar power plants presents a unique opportunity to combine the benefits of both technologies [26].

Currently, the utilization of ammonia as a fuel for gas turbines is primarily centered on its combination with other gases such as

* Correspondence to: Department of Mechanical Engineering (Robotics) and MATEC, Guangdong Technion–Israel Institute of Technology, 241 Daxue Road, Shantou, Guangdong, 515063, China.

E-mail address: dmitry.pashchenko@gtiit.edu.cn.

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hydrogen, methane, or synthesis gas [27–29]. This approach is adopted due to the inherent limitations of pure ammonia, including its low combustion efficiency and stability [30–32]. Additionally, the slow flame speed of ammonia requires a larger combustion volume that demands modifications to the combustion chambers of conventional gas turbines [33,34]. By blending ammonia with hydrogen or methane, it becomes feasible to leverage the existing infrastructure of gas turbines and enhance the overall performance and efficiency of the combustion process [35,36].

Alikulov et al. studied various blends of natural gas (NG), ammonia and hydrogen on the base of Fergana Combined Heat and Power plant [37]. The authors reported that the fuel blends with ammonia and hydrogen could be combusted using the existing infrastructure. It was observed that a fuel blend of 40% vol. NG, 30% of H_2 and 30% of NH_3 yielded the highest power output and lowest CO_2 emissions. However, the production of H_2 and NH_3 fuels required high costs, which resulted in a high total cost per kWh of power output.

Aalrebei et al. [38] compared the efficiency of the gas turbine cycle using 70%NH_3/30%H_2 fuel blend and CH_4. The combustion of NH_3/H_2 fuel blend produced more net turbine work and lower turbine temperatures for fuel-to-air equivalence ratios lower than 0.8. Also, the emission performance is also better for NH_3/H_2/air for an equivalence ratio lesser than 0.75. For the combustion equivalence ratio close to stoichiometric the CH_4/air combustion showed better performance.

Even a small H_2 addition to NH_3 fuel promotes the combustion characteristics. Wei et al. [39] investigated the addition of 10% mole fraction of H_2 to NH_3 fuel in a swirled gas turbine combustor. The H_2 addition promoted the flame stabilization due to higher flame speed and stronger stretch resistance. The combustion was further improved as H_2 oxidation has a high reaction rate and improves the ammonia oxidation as well.

The NH_3/CH_4 fuel blend influence on characteristics of the commercial micro gas turbine (mGT) Ansaldo AE-T100 was studied by Avila et al. [40]. The AE-T100 micro gas turbine was originally designed for natural gas combustion and NH_3 was added to view the possibility of fuel substitution. The authors reported a stable operation of mGT for the power output of 60 kWe and ammonia volume fraction less than 0.63. The increase in NH_3 mole fraction leads to lower CO_2 emissions, but much higher NO_x emissions. Therefore, the mGT hardware has to be modified to reduce the emissions and improve the performance.

Skabelund et al. [41] performed a thermodynamic analysis of a 50MW gas turbine with various fuel blends of $NH_3/H_2/CH_4$. The system operation efficiency was the highest (44.6%) for fuel blends with high ammonia concentration. An increase in ammonia mole fraction led to a higher fuel mass flow rate by 2 times on average. For the same fuel consumption as pure CH_4, the 22% NH_3 and 78% H_2 fuel blend increased the system efficiency by 3%.

Recently, ammonia has found great interest as a hydrogen carrier because it can be easily converted to hydrogen-rich gas via the chemical reaction [42,43]:

$$2NH_3 \longrightarrow N_2 + 3H_2; \Delta H_{298} = 46.19 \text{ kJ/mol.} \tag{1}$$

The ammonia thermochemical transformation (ATT) process is an endothermic process over catalysts, and external heat sources must be supplied to convert ammonia into hydrogen-rich gas [44,44]. Heat sources for ammonia thermochemical transformation include:

- electric heating, which provides a clean and efficient source of heat [45,46];
- ammonia combustion to provide heat for the thermochemical transformation process [47,48];
- exhaust heat from gas turbines with temperatures up to 500–600 °C, which can be used to supply external heat for the decomposition process [43,49,50];
- concentrated solar energy, which can provide the high temperatures required for ammonia thermochemical transformation [51, 52].

The utilization of renewable energy sources, particularly solar energy, for the thermochemical transformation of ammonia in gas turbines holds significant promise. Solar energy has emerged as a viable and sustainable option for powering various industrial processes, including the decomposition of ammonia (NH_3) into its constituent gases. In recent years, there has been remarkable progress in leveraging solar energy for power generation. Siddiqui [53] provided an analysis of scheme with utilized excess solar energy from a photovoltaic solar plant for ammonia production. The hydrogen required for ammonia production was obtained due to water electrolysis during the hours of the excess power output. The N_2 was captured using pressure swing adsorption. The produced ammonia was the means of chemical energy storage and was used as a fuel for the ammonia fuel cell during the hours of low energy availability. Yu et al. [54] proposed using a parabolic trough solar collector as a heat source for ammonia decomposition. The authors obtained practically complete NH_3 decomposition for moderate process temperatures and mixture residence time. The produced H_2 could be used for power generation.

One particularly promising approach involves harnessing concentrated solar energy to provide the high temperatures required for the thermochemical transformation of ammonia. Concentrated solar energy systems can generate heat sources with temperatures ranging up to 1200 °C, which are well-suited for driving the decomposition reactions involved in the conversion of ammonia into hydrogen-rich gas. By utilizing concentrated solar energy, it becomes possible to achieve the high temperatures necessary for efficient ammonia decomposition while simultaneously minimizing the reliance on fossil fuels and reducing greenhouse gas emissions.

By integrating ammonia-fed gas turbines and concentrated solar power plants, a hybrid power generation system can be developed that leverages the strengths of each technology while mitigating their respective limitations. This paper explores the potential synergies, challenges, and opportunities associated with the integration of ammonia-fed gas turbines and concentrated solar power plants for enhanced CO_2-free power generation. There is a lack of knowledge about the integrated ammonia-fed gas turbines and concentrated solar power plants, particularly, the schematic diagram, fuel and energy consumption, feasibility analysis. To that end, we have conducted the thermodynamic analysis of the novel integrated solar ammonia-fed gas turbines to understand the effect of operational parameters on the thermal efficiency. Moreover, we performed a comparative analysis of the combustion performance of ammonia and blends derived from ammonia thermochemical transformation to understand the possibility of using existing gas turbines designed for natural gas combustion for integrated solar combined cycle system fueled with ammonia.

2. Methodology

2.1. Concept

Ammonia has garnered attention as a potential fuel source, but its utilization in combustion processes presents several challenges. These challenges include its low flammability, high NO_x emissions, and low radiation intensity. In the context of gas turbine applications, the primary obstacles are the low flammability and unstable combustion characteristics of ammonia (low flame speed, long quenching distance). To address these challenges, modifications to the combustion chamber are necessary to increase the combustion volume and ensure stable operation. Consequently, existing gas turbine systems must undergo significant modifications to accommodate the use of ammonia as a fuel.

On the other hand, hydrogen stands out among popular gas turbine fuels due to its high flame speed and stable combustion properties. When comparing the combustion characteristics of various gas turbine fuels, hydrogen exhibits superior performance.

Furthermore, the production of hydrogen for an ammonia-hydrogen gas turbine fuel blend can be achieved through on-board generation

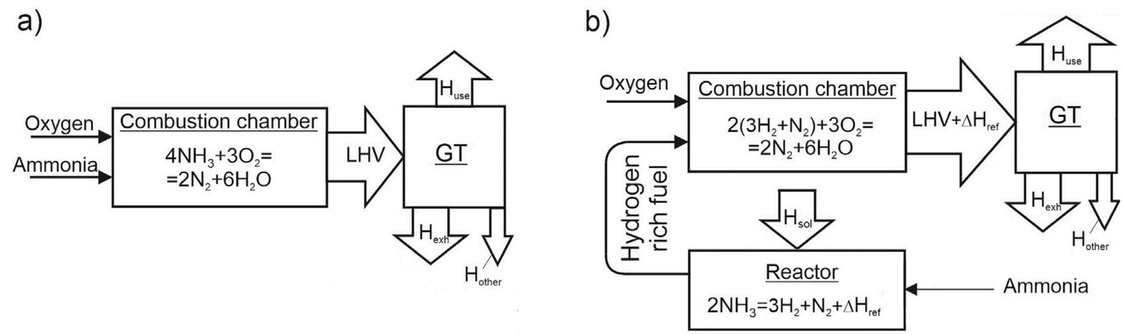

Fig. 1. Conversion of chemical energy of ammonia into heat: (a) traditional ammonia-fed gas turbine; (b) integrated solar ammonia-fed gas turbine.

from ammonia via thermochemical decomposition. This approach offers a pathway to leverage the advantageous combustion properties of hydrogen while utilizing ammonia as a sustainable fuel source. By combining these two fuels, it is possible to enhance the overall performance and efficiency of gas turbine systems, paving the way for cleaner and more environmentally friendly energy generation processes.

In summary, the integration of hydrogen into an ammonia-based fuel blend for gas turbines holds significant potential for overcoming the challenges associated with ammonia combustion. Through innovative solutions such as on-board hydrogen production from ammonia, it becomes feasible to enhance combustion characteristics, improve efficiency, and reduce environmental impact in gas turbine applications.

The concept of integrated solar ammonia-fed gas turbine is based on the principles of thermochemical ammonia transformation before combustion using concentrated solar energy. In this case, solar energy is converted into the chemical energy of the products of ammonia decomposition. In other words, the concept of a chemical pump is realized in such gas turbines [55,56]. Schematically, the conversion of chemical energy of ammonia into heat can be divided into two virtual stages as presented in Fig. 1b. The first stage is the thermochemical transformation of primary ammonia into hydrogen-rich gas. This stage can be named the “chemical pump” because here the low potential solar energy is converted into the chemical energy of hydrogen-rich gas. The second stage is the combustion of a new hydrogen-rich fuel. At the same time, the conversion of chemical energy of ammonia into heat for traditional gas turbines is taking place in one stage — direct combustion. The comparison of the conversion of chemical energy into heat for traditional gas turbines (a) and integrated solar ammonia-fed gas turbine (b) is presented in Fig. 1.

The schematic diagram depicted in Fig. 2 illustrates the integrated solar ammonia-fed gas turbine system with a steam turbine. This system comprises three main subsystems: the ammonia reforming with hydrogen-rich storage, the gas turbine, and the steam turbine cycle. Within this scheme, ammonia undergoes a chemical reaction in the ammonia reformer (1), which is heated by solar energy, resulting in the production of hydrogen-rich gas as a product. To address the intermittency of solar radiation and ensure continuous operation, a hydrogen-rich gas storage system is incorporated into the scheme. The hydrogen-rich gas generated by the reformer is stored in a tank and later utilized as fuel for the gas turbine. Additionally, the exhaust heat from the gas turbine is used in a steam turbine cycle to further enhance energy efficiency. This integrated system effectively leverages solar energy to produce hydrogen-rich gas for power generation, thereby optimizing energy utilization and sustainability.

The schematic diagram of the CO2-free power plant offers several key advantages that contribute to its overall efficiency and sustainability. Firstly, the utilization of hydrogen-rich gas as a fuel for the gas turbine enables more stable and efficient combustion compared to traditional fossil fuels. This not only reduces emissions but also enhances the overall performance of the system.

Moreover, the integration of low-potential solar energy into a high-temperature gas turbine cycle represents a significant advancement in energy utilization. By harnessing solar energy at higher temperatures, well beyond the typical 500 °C limit of conventional solar power cycles, the system achieves notably higher efficiency levels. This innovative approach maximizes the conversion of solar energy into usable power, making the system more effective and environmentally friendly.

Another key advantage of this integrated system is its ability to address the challenge of uneven solar radiation throughout the day. By storing hydrogen-rich gas produced during peak sunlight hours, the system ensures a continuous and reliable fuel supply for the gas turbine, even when solar energy availability fluctuates. This feature enhances the system’s reliability and stability, making it a viable option for consistent power generation.

Furthermore, the design allows for the adaptation of existing gas turbines to utilize ammonia as a fuel source. This flexibility enables the retrofitting of conventional gas turbine systems, reducing the need for extensive modifications or new equipment. By leveraging existing infrastructure, the system becomes more cost-effective and accessible for implementation in various settings.

2.2. Thermodynamic analysis

A thermodynamic analysis was conducted to investigate the effect of various operational parameters on the thermal efficiency of novel system and to assess the income and outcome of energy balance. This analysis involved studying the relationship between different operating conditions, such as temperature, pressure, and flow rates, on the overall efficiency of the system. The thermodynamic analysis conducted in this research utilized Aspen HYSYS [57,58], a software program renowned for its capabilities in simulating and analyzing gas and steam turbine cycles, as well as ammonia decomposition process [59,60].

Specifically, the thermodynamic analysis involved two main components: cycle analysis and ammonia decomposition system analysis. The cycle analysis focused on evaluating the efficiency and performance of gas and steam turbine cycles under various operating conditions, such as temperature, pressure, and flow rates. On the other hand, the ammonia decomposition system analysis focused on studying hydrogen-rich gas composition and enthalpy of process. A computational scheme of integrated solar ammonia-fed gas turbine with bottom steam turbine for thermodynamic analysis employed via Aspen HYSYS presented in Fig. 3a.

In addition, in this study we analyzed the simple integrated solar combined cycle power plant fed with ammonia without thermochemical ammonia decomposition [61]. In such system, the solar energy is used to produce steam for steam turbine cycle as shown in Fig. 3b. In the computational schemes from Fig. 3, solar energy is equal as well as other operational and design parameters of steam and gas turbine for both schemes.

The computational scheme in Fig. 3a shows that ammonia is first directed to a reformer before entering the combustion chamber. In

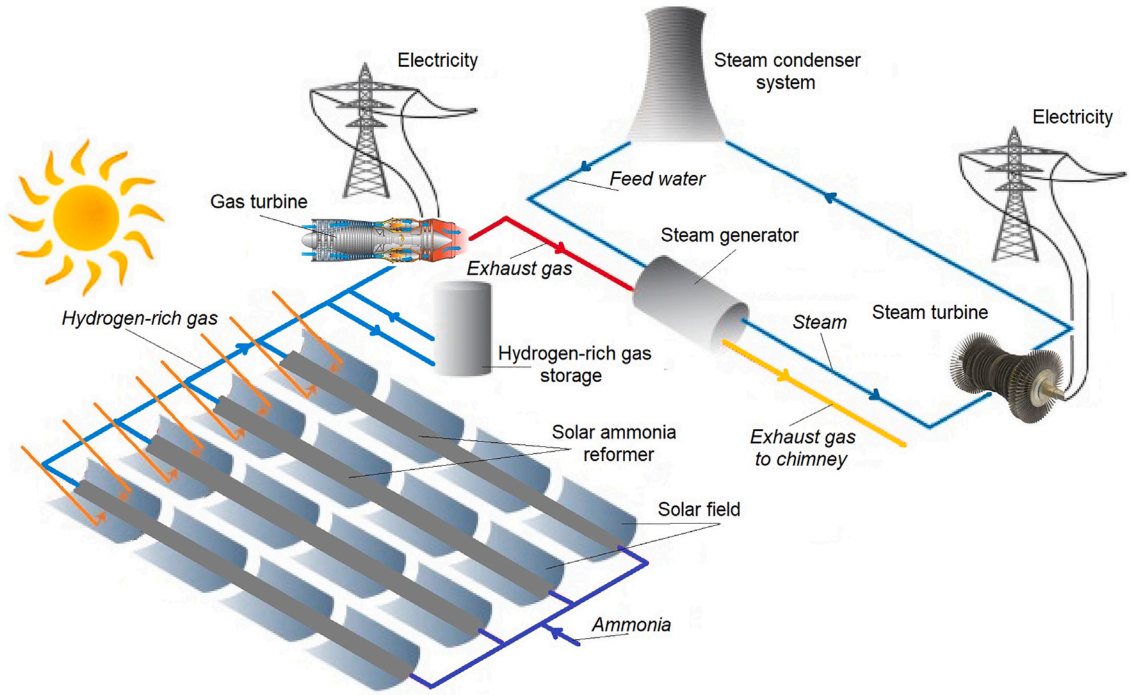

Fig. 2. Schematic diagram of integrated solar ammonia-fed gas turbine with bottom steam turbine.

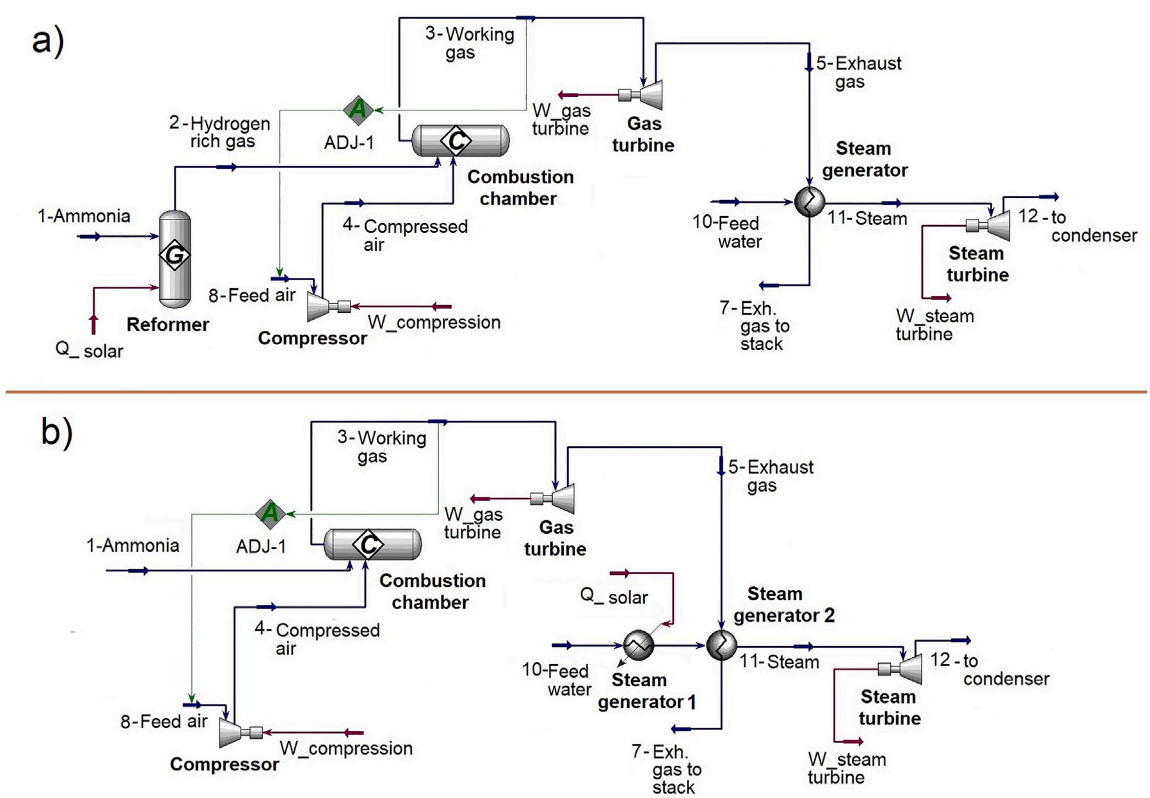

Fig. 3. Computational scheme of integrated solar ammonia-fed gas turbine with bottom steam turbine for thermodynamic analysis employed via Aspen HYSYS: (a) with thermochemical ammonia decomposition using solar energy; (b) without thermochemical ammonia decomposition (solar energy is used for steam generation).

the reformer, solar energy (Q_{solar}) is supplied to both counteract the endothermic effect of reaction (1) and raise the temperature of the reaction mixture to that of the hydrogen-rich gas at point 2. The resulting hydrogen-rich gas exiting the reformer is then mixed with compressed air from the compressor and fed into the combustion chamber, where it produces the hot working gas. This high-temperature gas drives the gas turbine to generate mechanical work ($W_{gasturbine}$), while the exhaust heat is recovered to produce steam for the bottoming cycle.

For comparison, Fig. 3b depicts the computational scheme of a conventional integrated solar combined cycle power plant. In this

Table 1
The operational and design parameters of ammonia fed CCPP with CRGT.

Parameter	Value
Turbine inlet temperature (TIT)	800–1400 °C
Compression/expansion ratio	5–25
Turbine isentropic efficiency (both ST and GT)	90%
Compressor isentropic efficiency	85%
Steam temperature at steam turbine inlet	300 °C
Steam pressure	20 bar
Condenser pressure	0.06 bar
Exhaust gas temperature after steam generator	110 °C
Ambient temperature	20 °C
Ambient pressure	1.013 bar
Ammonia temperature	20°
Inlet air/combustion chamber pressure loss	0.5% / 3.0%
Pressure loss on gas/water/steam sides	1.3% / 4.8% / 3.1%

Table 2
Ammonia molar fraction in the products of decomposition reaction.

Temperature, °C	Source	Pressure, bar			
		10	30	50	100
350	Aspen HYSYS	7.29	17.71	24.98	–
	Larson&Dodge [64]	7.35	17.80	25.11	–
375	Aspen HYSYS	5.21	13.29	19.34	30.78
	Larson&Dodge [64]	5.25	13.35	19.44	30.95
400	Aspen HYSYS	3.81	10.01	15.06	24.78
	Larson&Dodge [64]	3.85	10.09	15.11	24.91
425	Aspen HYSYS	2.78	7.56	11.67	20.15
	Larson&Dodge [64]	2.80	7.59	11.71	20.23
450	Aspen HYSYS	2.01	5.76	9.12	16.29
	Larson&Dodge [64]	2.04	5.80	9.17	16.35
475	Aspen HYSYS	1.58	4.49	7.09	12.95
	Larson&Dodge [64]	1.61	4.53	7.13	12.98
500	Aspen HYSYS	1.18	3.43	5.52	10.35
	Larson&Dodge [64]	1.20	3.48	5.58	10.40

configuration, ammonia is supplied directly to the combustion chamber of the gas turbine, while solar energy is exclusively utilized for steam generation in the bottoming cycle.

The thermodynamic analysis of integrated solar ammonia-fed gas turbine with bottom steam turbine cycle is focused on the calculation of energy inputs and outputs in the system. Overall work output of such system is given by

$$W_{sys} = W_{gt} + W_{st} - W_{comp} \tag{2}$$

where W_{gt}, W_{st} is a work output in a gas turbine and steam turbine; W_{comp} is a work input in a compressor (work input for pump in steam turbine cycle assumed as negligible).

The thermal efficiency of the analyzed system can be written as follows [62]:

$$\eta_{sys} = \frac{W_{sys}}{\dot{m}_{\mathrm{NH_3}} \cdot LHV_{\mathrm{NH_3}} + Q_{sol}} \tag{3}$$

where W_{sys} is a overall work output; $\dot{m}_{\mathrm{NH_3}}$ is an ammonia mass flow rate; $LHV_{\mathrm{NH_3}}$ is a lower heating value of ammonia; Q_{sol} is solar energy supplied to system.

Solar energy is used to compensate the endothermic effect of reaction (1) and to heat the reaction mixture [63]:

$$Q_{sol} = \Delta H_{sol} = \Delta H_{298} + \Delta H_{mix} \tag{4}$$

where ΔH_{298} – the reaction enthalpy; ΔH_{mix} – change of enthalpy of reaction mixture which can be determined as difference between the enthalpy of hydrogen rich gas after reformer and the enthalpy of ammonia before reformer.

The initial parameters of the integrated solar combined cycle power system fueled with ammonia is provided in Table 1.

The second part of the thermodynamic analysis is dealing with ammonia decomposition process. The thermodynamic analysis of the ammonia decomposition process involves the use of Gibbs free energy minimization method to determine the equilibrium composition of the reaction mixture [65,66]. The reaction is assumed to take place in a closed system at constant temperature and pressure. The reaction is represented by Eq. (1), and the Gibbs free energy change (ΔG) for this reaction is calculated using thermodynamic data. The equilibrium composition of the reaction mixture is then calculated using the mole balance equations and the knowledge of the initial composition of the mixture. Peng-Robinson equation is used as an equation of state for the analyzed system [67]. The effect of temperature and pressure on the equilibrium composition is also studied by varying these parameters and calculating the corresponding equilibrium compositions. The thermodynamic analysis provides insight into the feasibility and conditions required for the ammonia decomposition process to occur.

The thermodynamic analysis of ammonia decomposition process is performed via Gibbs free energy minimization method which is based on the principles that the analyzed chemical system is thermodynamically favorable when total Gibbs free energy has lowest value and differential of Gibbs free energy is zero for a given operation conditions [68]:

$$dG^t = 0 \tag{5}$$

where G^t is total Gibbs free energy.

For the moderate low pressure and high temperature, the ammonia decomposition system can be assumed as ideal [69].

Total Gibbs free energy of the ammonia decomposition system can be calculated as a sum of chemical potentials of all species:

$$G^t = \sum n_i \mu_i \tag{6}$$

where n_i is number of moles of species i; μ_i is chemical potential of species i.

The chemical potentials of ith element can be calculated using the following equation:

$$\mu_i = \Delta G^0_{fi} + RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{7}$$

where G^0_{fi} is standard Gibbs free energy of ith species formation; R is molar gas constant; T is temperature of the system; f_i is fugacity of pure species i; f_i^0 is fugacity of pure species i at the standard state.

Combining the equations above, the equation for total Gibbs free energy of the ammonia decomposition system has the following form:

$$G^t = \sum n_i \Delta G^0_{fi} + \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{8}$$

The following assumptions were made for the thermodynamic analysis of the ammonia decomposition process:

- The system is assumed to be in a state of thermodynamic equilibrium, which means that the chemical reactions occur at a constant temperature and pressure [70];
- The ammonia decomposition reaction is assumed to be a reversible reaction, which means that the forward and reverse reactions occur simultaneously and at the same rate [51,71];
- The reaction products are assumed to be pure gases, which means that there are no impurities or other by-products that may affect the thermodynamic analysis; the following species are involved in the analysis: NH_3, N_2, H_2 [72];
- The reaction is assumed to occur in an adiabatic reactor, which means that there is no heat transfer between the reactor and its surroundings [73];
- The pressure drop across the reactor is assumed to be negligible [73];
- The reactor is assumed to be well-mixed, which means that the reactants and products are uniformly distributed throughout the reactor [73].

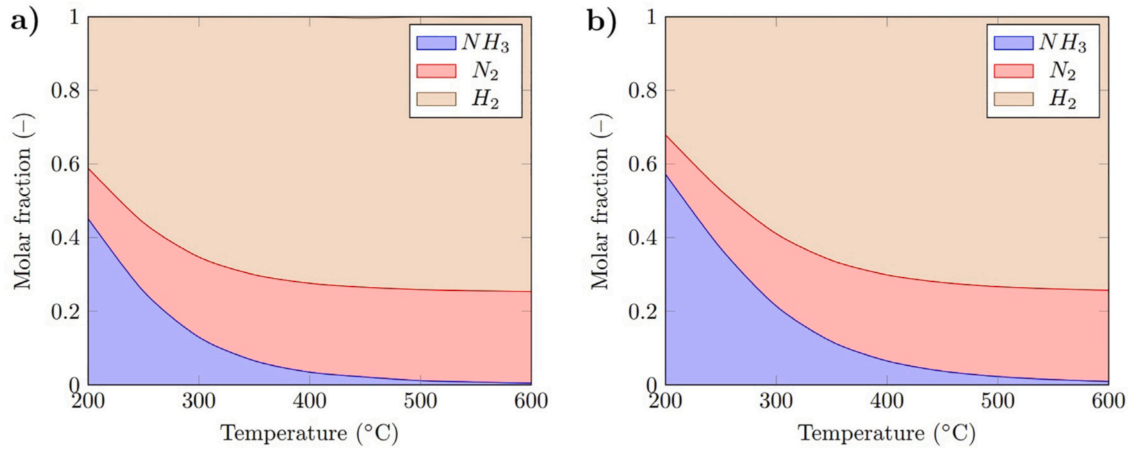

Fig. 4. The composition of the product gas of ammonia decomposition reaction: (a) p = 10 bar; (b) p = 20 bar.

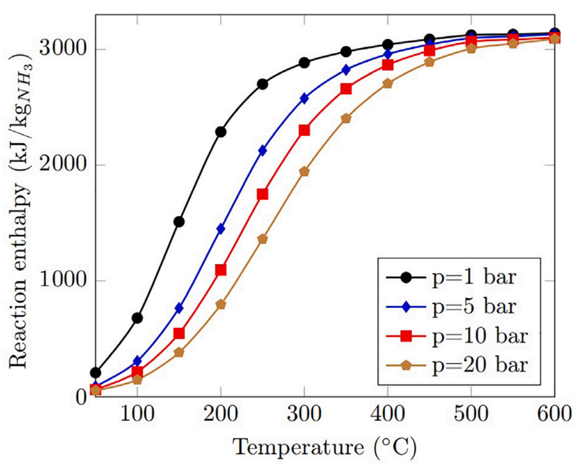

Fig. 5. Reaction enthalpy as a function of temperature and pressure.

Moreover, in the analysis, we assume that the solar heat supplied to the reformer is equal to the enthalpy of the ammonia decomposition process that includes the enthalpy of reaction (1) and change of reaction mixture enthalpy. Therefore, the heat flow with solar energy depends on the temperature. We analyzed a wide range of temperature regimes.

The study acknowledges three key practical limitations of solar thermal supply for ammonia decomposition:

1. Diurnal variability: The solar flux cannot continuously match the reaction's thermal demand (600–800 °C), requiring either:
   - Thermal energy storage (as demonstrated by Dunn et al. using molten salts [74]);
   - Hybrid solar-fossil systems (per Huo et al. [75]).
2. Conversion efficiency limits: Even with optimal catalysts, solar-to-hydrogen efficiencies typically reach 40%–55% due to:
   - Re-radiation losses at high temperatures;
   - Incomplete ammonia conversion (<99%) below 500 °C.
3. Scalability challenges: Large-scale plants require:
   - High-precision solar tracking systems;
   - Advanced cavity receivers to maintain temperature uniformity.

3. Results

3.1. Verification

The equilibrium composition of products from the ammonia decomposition process was calculated using Aspen HYSYS. To verify the results, the ammonia compositions in the product gas were compared for a wide range of operating parameters, including pressure and temperature. The reference values for ammonia molar fraction in the product gas were obtained from experimental results obtained at the Fixed Nitrogen Research Laboratory (F.N.R.L) [64]. Table 2 shows the calculated ammonia molar fraction using Aspen HYSYS and the values obtained from F.N.R.L [64]. The results show a deviation of less than 0.5% for all operational parameters, indicating that Aspen HYSYS provides a good correlation with experimental data for the equilibrium gas composition. The data from Table 2 show a good correlation between the reference value and calculated results obtained via Aspen HYSYS.

The ammonia decomposition reaction is an endothermic reaction, and the number of moles of products is higher than the number of moles of the initial reagent. The Le Chatelier principle, a fundamental concept in chemical equilibrium, dictates that changes in temperature and pressure can influence the direction of a chemical reaction. For an endothermic reaction of ammonia decomposition, where heat is absorbed, increasing the temperature favors the formation of products by shifting the equilibrium towards the right. Conversely, an increase in pressure tends to suppress the formation of products.

Applying this principle to the decomposition of ammonia, it can be concluded that increasing the temperature and decreasing the pressure would promote the breakdown of ammonia molecules into their constituent elements. This is substantiated by the analysis of the product composition depicted in Fig. 4. The data presented in Fig. 4 illustrates that at temperatures above 500 °C and pressures within the range of 10 to 20 bar, the presence of unreacted ammonia is minimal, indicating complete conversion to a hydrogen-rich gas. In fact, hydrogen constitutes approximately 75% of the resulting gas mixture under these conditions.

The findings derived from Fig. 4 underscore the necessity of operating at temperatures above 500 °C to generate a gas stream enriched with hydrogen, with concentrations reaching up to 75% from the decomposition of ammonia. Notably, even at a lower temperature of 200 °C, a significant fraction of hydrogen is produced. Achieving and maintaining such elevated temperatures is attainable in contemporary solar power facilities, showcasing the potential for leveraging this process in sustainable energy production [19,76].

The degree of ammonia decomposition determines the enthalpy of the reaction, which also depends on temperature and pressure.

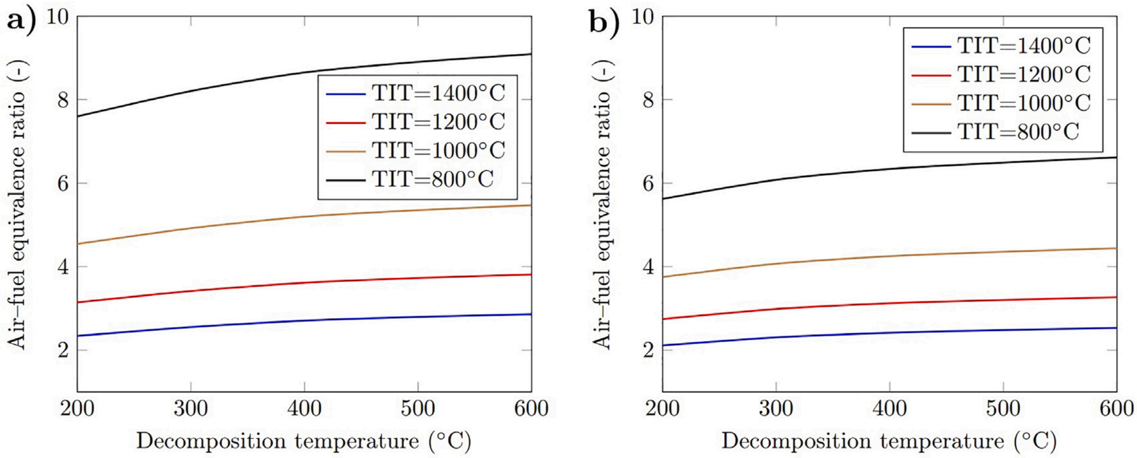

Fig. 6. Air-fuel equivalence ratio in ammonia fed solar integrated gas turbine: (a) p = 10 bar; (b) p = 20 bar.

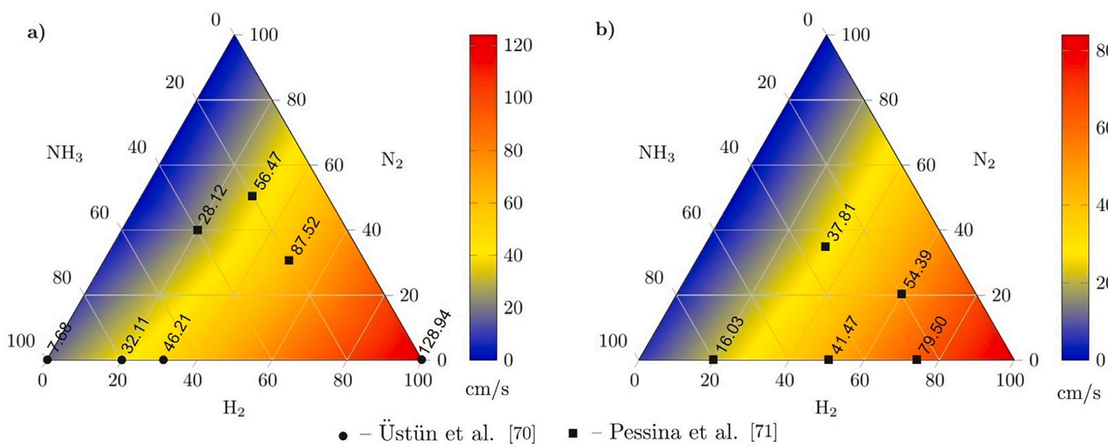

Fig. 7. Laminar flame speed for NH_3:H_2:N_2 blends: (a) p = 10 bar; (b) p = 20 bar.

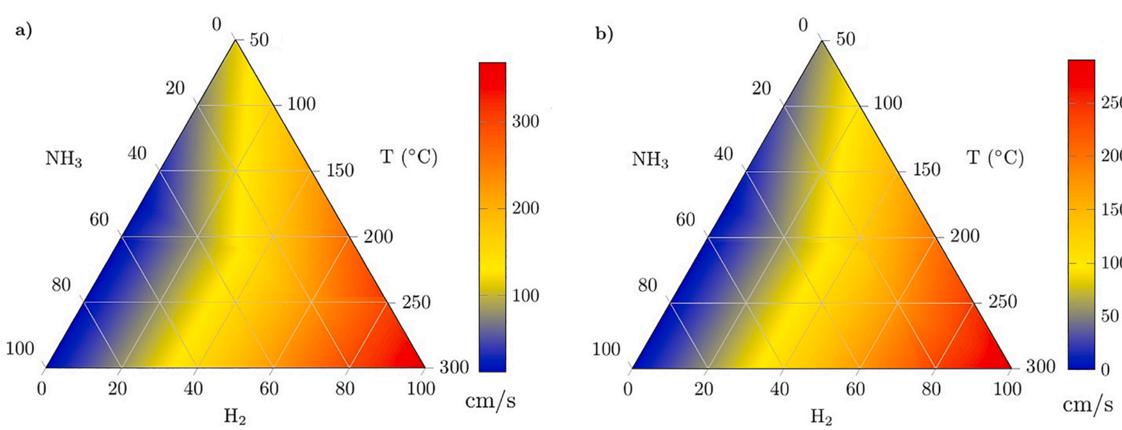

Fig. 8. Laminar flame speed for ammonia-hydrogen blends vs. temperature: (a) p = 10 bar; (b) p = 20 bar.

On the other hand, the reaction enthalpy shows the amount of solar heat required to obtain an equilibrium product gas composition and determines the amount of solar heat converted into chemical energy of a hydrogen-rich gas. Therefore, the combustion heat of the products is higher than the combustion heat of ammonia. Fig. 5 shows the effect of temperature and pressure on the reaction enthalpy. Fig. 5 reveals that 1 kg of ammonia can absorb about 3100 kJ when the reaction temperature is above 500 °C, and the combustion heat of a hydrogen-rich gas is higher than the combustion heat of ammonia by 3100 kJ/kg_{NH_3}.

3.2. Air–fuel equivalence ratio

According to the second law, the thermal efficiency of a gas turbine increases when the turbine inlet temperature increases. The combustion temperature of ammonia and hydrogen-rich gas produced from ammonia decomposition can be higher than 2000 °C. However, the gas turbine inlet temperature is limited by the thermal resistance of its elements. Therefore, the turbine inlet temperature is lower than the adiabatic temperature. The control of the turbine inlet temperature can be accomplished via a change in air mass flow rate. A higher air mass flow rate at a constant fuel mass flow rate gives a lower turbine inlet temperature. In this case, the air-to-fuel ratio is higher than stoichiometric. An increase in the air-fuel equivalence ratio leads to a decrease in laminar flame speed.

Moreover, in ammonia-fed solar integrated gas turbines, an additional portion of heat is supplied to the thermodynamic cycle with hydrogen-rich gas. This additional heat leads to an increased air flow rate to control the turbine inlet temperature. Fig. 6 shows the air-fuel equivalence ratios in ammonia-fed solar integrated gas turbines for various operational parameters.

Fig. 6 shows that an increase in the temperature of the ammonia decomposition reaction results in an increase in the air-fuel equivalence ratio because the enthalpy of the reaction is a function of temperature. When the temperature of the hydrogen-rich gas increases, additional heat is supplied to the combustion chamber of the gas turbine. Thus, to control the turbine inlet temperature, the air mass flow rate should be increased, which leads to an increase in the air-fuel equivalence ratio. An increase in the air-fuel equivalence ratio leads to a decrease in the fuel-air ratio and, as a result, a decrease in laminar flame speed. For example, the air-fuel equivalence ratio is 3–4 times higher for a turbine inlet temperature of 800 °C than for a turbine inlet temperature of 1400 °C. Consequently, an increase in the air-fuel equivalence ratio results in a lower laminar flame speed, and more space in the combustion chamber is required for complete combustion of the hydrogen-rich gas.

3.3. Laminar flame speed

One of the main challenges with using ammonia in existing gas turbines is its low flame speed. This can result in incomplete combustion and reduced combustion efficiency. To address this issue, the combustion chamber volume needs to be increased to ensure complete combustion. Additionally, ammonia decomposition before combustion changes the gas turbine fuel composition. As shown in Fig. 4, at temperatures above 500 °C, the hydrogen-rich fuel primarily consists of hydrogen, which has a significantly higher flame speed than ammonia.

Flame speed is influenced by various factors such as fuel composition, initial temperature, and pressure. The impact of initial temperature on the combustion of hydrogen-rich gas is crucial for the operation of ammonia-fed solar integrated gas turbines, as the temperature of the hydrogen-rich gas can reach 600–700 °C after ammonia decomposition. Higher temperatures generally result in increased flame speeds, while higher pressures tend to decrease flame speed. Fig. 7 illustrates the laminar flame speeds for different compositions of ammonia-hydrogen-nitrogen mixtures obtained from Chemkin-Pro, a powerful tool for data obtaining on flame speeds. The results from Chemkin-Pro are compared with data from other studies [77,78], showing good correlation. Even mixtures with low H_2 concentrations exhibit suitable flame speeds for gas turbines. Furthermore, Fig. 7 demonstrates the impact of pressure on flame speed, indicating that higher pressures lead to lower flame speeds.

According to Mallard's the theory, the laminar flame speed is a functions of gas blend properties such as the thermal diffusivity of the gas blend, the combustion reaction rate and the adiabatic temperature:

$$s_L^\circ = \sqrt{\alpha\dot{\omega}\left(\frac{T_b - T_i}{T_i - T_u}\right)} \tag{9}$$

where α is thermal diffusivity; $\dot{\omega}$ is reaction rate; and the temperature subscript u is for unburned, b is for burned and i is for ignition temperature.

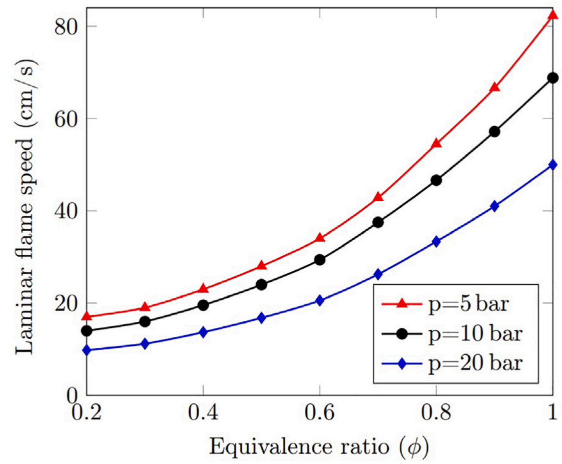

Fig. 9. Laminar flame speed of hydrogen-rich gas with H_2:N_2 = 3:1 as a function of fuel-air equivalence ratio.

An increase in the temperature of hydrogen-rich gas results in a corresponding rise in adiabatic temperature, leading to an increase in flame speed. This relationship between initial blend temperature and laminar flame speed is depicted in Fig. 8 for pressures of 10 bar and 20 bar. The graph illustrates that as the initial blend temperature rises, the laminar flame speed also increases. For instance, at an initial blend temperature of 300 °C, the laminar flame speed for a NH_3:H_2 blend of 1:9 at 10 bar pressure is approximately 318.12 cm/s, compared to 143 cm/s at 100 °C. These results are well correlated with the results obtained by Ustun et al. [77]. This trend is consistent for blends with hydrogen mole fractions exceeding about 60%. However, for blends with higher ammonia molar fractions, the impact of temperature on laminar speed is less pronounced, indicating that the benefits of temperature increase are more significant for hydrogen-rich gas.

Ammonia-fed gas turbines typically operate under high pressures ranging from 10 to 25 bar. Fig. 8 further demonstrates the influence of pressure on laminar flame speed. As depicted in the figure, pressure has a noticeable effect on hydrogen-rich blends, with laminar flame speeds significantly higher than those for ammonia-rich mixtures.

The regulation of turbine inlet temperature in gas turbines is achieved by adjusting the air mass flow rate, where a higher turbine inlet temperature results in a lower mass flow rate of air passing through the compressor. This relationship is illustrated in Fig. 6, which shows that the air mass flow rate in gas turbines is typically higher than stoichiometric conditions. In the rich combustion zone, the air-fuel equivalence ratio, as well as the fuel-air equivalence ratio, plays a crucial role in determining the laminar flame speed.

Fig. 9 demonstrates the impact of the fuel-air equivalence ratio (ϕ) on the laminar flame speed for a hydrogen-rich fuel mixture with an H_2:N_2 ratio of 3:1. The graph reveals that the laminar flame speed decreases non-monotonically as the equivalence ratio decreases. Specifically, at stoichiometric equivalence ratio, there is a significant deviation in laminar flame speed for different pressures, whereas at ϕ=0.2, this deviation is considerably reduced in absolute terms. Therefore, it can be concluded that laminar flame speed of hydrogen rich fuel for gas turbine conditions has an appropriate value comparing to low laminar flame speed of pure ammonia.

3.4. Thermal efficiency and heat balance

The utilization of solar energy for ammonia thermochemical decomposition in gas turbines and combined cycle power plants can enhance thermal efficiency. A detailed thermodynamic analysis, as outlined in the computational schemes depicted in Fig. 3, was conducted to

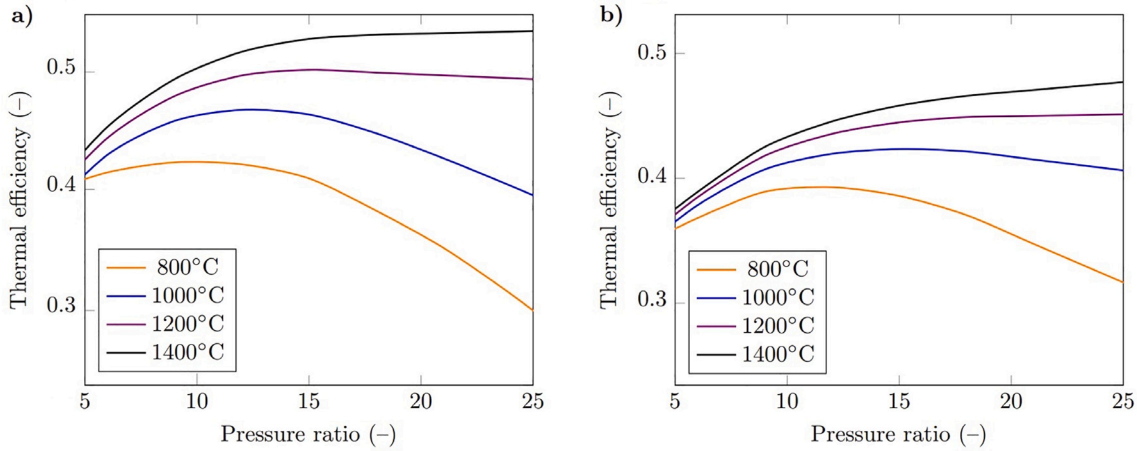

Fig. 10. Thermal efficiency of combined cycle power plant: (a) with ammonia thermochemical decomposition; (b) without ammonia thermochemical decomposition.

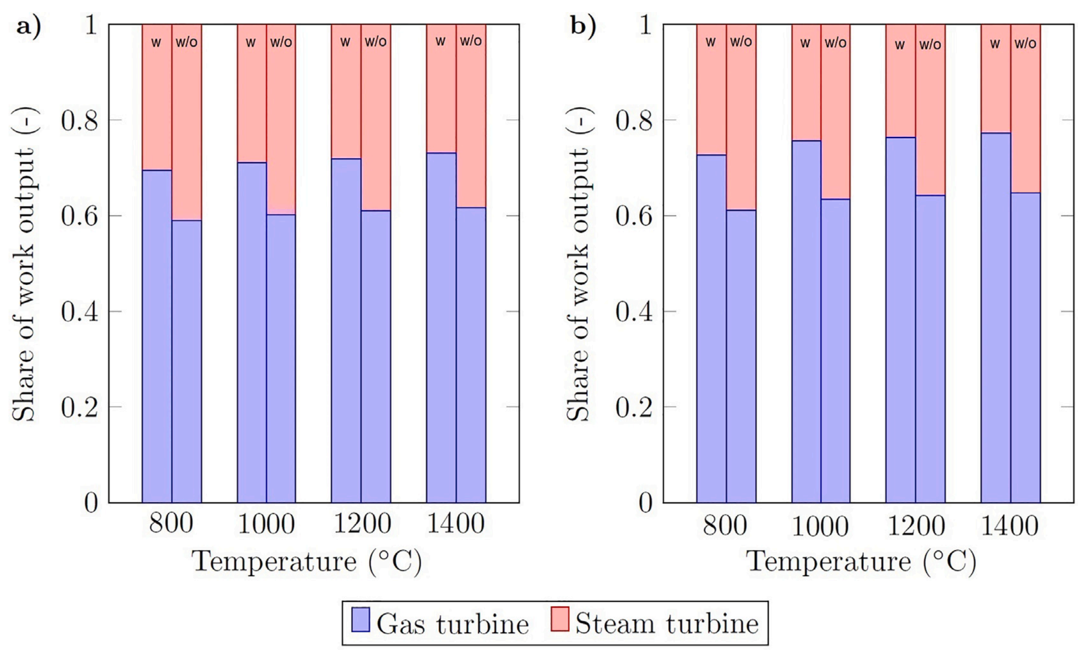

Fig. 11. Share of work output produced in gas turbine and steam turbine cycles for the schemes with ammonia thermochemical decomposition (w) and without ammonia thermochemical decomposition (w/o): (a) p = 10 bar; (b) p = 20 bar.

evaluate the effect of solar energy utilization on thermal efficiency and heat balance. It is expected that the thermal efficiency of combined cycles utilizing ammonia-fed gas turbines with thermochemical ammonia decomposition (Fig. 3a) will be higher that of conventional integrated solar combined cycles (Fig. 3b). This expectation stems from the fact that solar energy is integrated into the gas turbine cycle, which inherently boasts higher thermal efficiency compared to the steam turbine cycle.

Fig. 10 illustrates the thermal efficiency as a function of pressure ratio in the gas turbine for turbine inlet temperatures ranging from 800 to 1400 °C. The results are presented for both computational schemes outlined in Fig. 3: with and without thermochemical ammonia decomposition. The data from in Fig. 10 demonstrates that the use of solar energy for ammonia decomposition leads to an enhancement in thermal efficiency. This improvement is attributed to the utilization of solar energy in the form of chemical energy (combustion heat) within the gas turbine cycle, which inherently shows higher thermal efficiency compared to the steam turbine cycle. The observed increase in efficiency can be up to 3.2%.

Furthermore, Fig. 10 depicts the results obtained for a decomposition temperature of 500 °C. It is noted that an increase in decomposition temperature results in a higher proportion of solar heat being converted into the chemical energy of a new hydrogen-rich gas. Consequently, a higher decomposition temperature correlates with increased thermal efficiency due to more efficient conversion of solar heat.

According to the first law, ammonia thermochemical decomposition absorbs solar energy in the form of increase combustion heat as follows:

$$LHV_{H_2} = LHV_{NH_3} + \Delta H_{sol} \tag{10}$$

where LHV_{H_2}, LHV_{NH_3} – lower heating value of hydrogen-rich gas after ammonia decomposition and lower heating value of ammonia; ΔH_{sol} – solar energy absorbed during the ammonia decomposition process.

In the proposed concept of synergy of ammonia-fed gas turbines and concentrated solar power plants, solar energy is harnessed within a gas turbine cycle. The distribution of work output in gas turbine and steam turbine cycles is illustrated in Fig. 11 for systems with and without

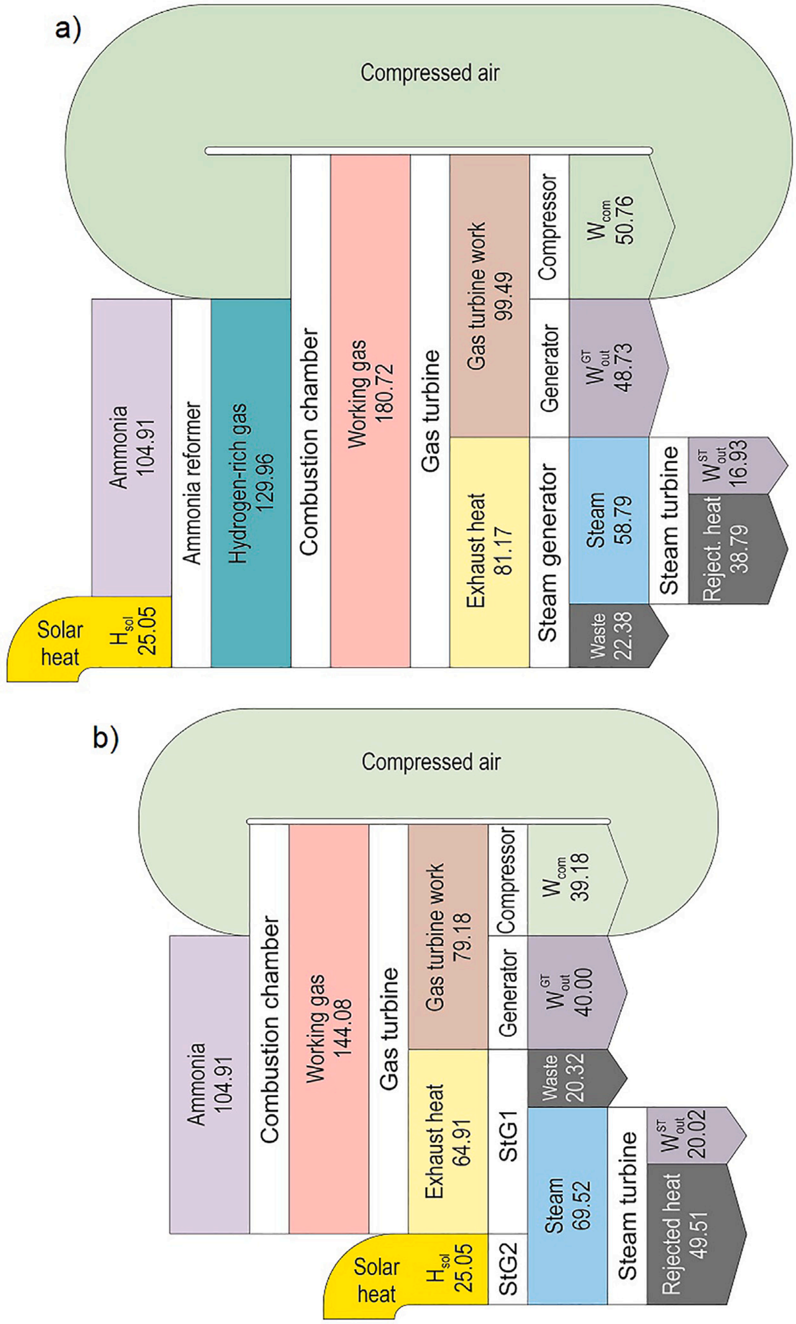

Fig. 12. Sankey diagrams for combined cycle power plane with (a) and without ammonia thermochemical decomposition (b): TIT = 1260 °C;p = 17.9 bar; normal work output of gas turbine of 40.00 MW (scheme b); ammonia decomposition temperature of 559 °C (All energy flows are in MW).

ammonia thermochemical decomposition. The results presented in Fig. 11 validate the previously mentioned expectation in Introduction and Methodology, indicating that in a combined cycle power plant (CCPP) utilizing ammonia decomposition with solar energy, a larger proportion of work is generated in the gas turbine compared to a CCPP without ammonia decomposition. Given the well-established higher thermal efficiency of gas turbine cycles over steam turbine cycles, the incorporation of ammonia decomposition leads to an increased overall thermal efficiency in CCPPs, as demonstrated in Fig. 10.

The distribution of work output shares is approximately 0.7:0.3 for a gas turbine pressure of 10 bar and around 0.75:0.25 for a pressure of 20 bar, respectively for the gas turbine and steam turbine cycles. Consequently, it can be inferred that using ammonia decomposition with solar energy in high-pressure gas turbines within CCPPs results in a greater proportion of work output being produced in the gas turbine cycle. This representation highlights that in CCPPs with ammonia decomposition, solar energy is utilized within the gas turbine cycle, whereas in CCPPs without ammonia decomposition, it is integrated into the steam turbine cycle.

To gain deeper insights into the heat flows within a combined cycle power plant with and without thermochemical ammonia decomposition, Sankey diagrams have been created, as depicted in Fig. 12. The gas turbine used as a reference in this study is Mitsubishi Power's ammonia-fed gas turbine from the H-25 series [35]. The Sankey diagrams were created based on specific operational parameters, including a turbine inlet temperature of 1260 °C, a pressure ratio of 17.9, a normal work output of 40.00 MW for the gas turbine, and an ammonia decomposition temperature of 559 °C.

The Sankey diagrams presented in Fig. 12 provide a visual representation of the distribution of heat flows within the ammonia-fed CCPP, highlighting key parameters such as the share of work output and thermal efficiency. By analyzing these diagrams, we can gain a comprehensive understanding of how heat is utilized and transferred within the combined cycle power plant under different operating conditions. The results derived from these Sankey diagrams contribute to elucidating the impact of thermochemical ammonia decomposition on the overall heat balance, performance and efficiency of the power generation system which based on combination of ammonia-fed gas turbines and solar power plants.

The Sankey diagrams from Fig. 12 show that for the same amount of input energy with ammonia and solar heat, an integrated solar combined cycle power plant with ammonia decomposition before combustion using solar energy generates 65.66 MW versus 60.04 MW from a combined cycle power plant with the use of solar energy for steam generation.

4. Conclusion

In this paper, a novel concept that explores the synergistic integration of ammonia-fed gas turbines and concentrated solar power plants was introduced. The core idea behind this concept involves utilizing solar energy to thermally decompose ammonia into hydrogen-rich fuel before combustion. This approach enhances combustion performance compared to using pure ammonia, as hydrogen-rich gas exhibits better combustion characteristics. Through our analysis, the laminar flame speed across a broad range of operational parameters, including initial temperature, pressure, fuel-air equivalence ratio, and hydrogen-rich gas composition was determined. The results indicate that higher temperatures and lower pressures lead to increased flame speeds, while reducing the fuel-air equivalence ratio tends to decrease the flame speed. Under gas turbine operating conditions, it was observed that laminar flame speeds ranging from 20 to 30 cm/s for a N_2:H_2 = 1:3 blend, representing complete ammonia decomposition into hydrogen-rich gas. The integration of ammonia-fed gas turbines and concentrated solar power plants yields a higher thermal efficiency compared to using each system independently, with thermal efficiency improvements of up to 3.5%. This enhancement is primarily attributed to a greater proportion of supplied heat being utilized in the gas turbine cycle. Specifically, in combined cycle power plants with ammonia thermochemical decomposition, approximately 75% of the work output is generated in the gas turbine cycle, while in systems without ammonia decomposition, this ratio is around 60%. The thermodynamic and flame speed analyses underscore the benefits of synergizing ammonia-fed gas turbines and concentrated solar power plants, particularly in terms of achieving higher thermal and combustion efficiency. Moreover, the proposed concept addresses the challenge of solar radiation intermittency by incorporating a hydrogen-rich gas storage system to ensure continuous operation. Moving forward, further investigation into emission performance and combustion stability will be crucial for fully understanding the potential advantages of integrating ammonia-fed gas turbines with concentrated solar power plants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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